

Netflix Titles Dataset - Exploratory Data Analysis Report

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Project: Exploratory Data Analysis (EDA)

Abstract

This project analyzes the Netflix Titles dataset using Exploratory Data Analysis (EDA). Data preprocessing techniques such as handling missing values, removing duplicates, and converting data types were applied before visualization. Different statistical graphs were used to identify trends, distributions, and relationships among variables. The analysis provides useful insights into Netflix's content library, release patterns, genres, ratings, and countries.

Objectives

- Understand dataset structure.
- Clean and preprocess the data.
- Visualize distributions and relationships.
- Identify trends and generate business insights.

Graph Explanations

Count Plot

Shows the distribution of Movies and TV Shows in the dataset. It compares the frequency of each content type. The visualization indicates that Movies are available in much larger numbers than TV Shows. This helps understand Netflix's content strategy. It provides a quick overview of the platform's catalog.

Bar Chart – Top Content Ratings

Displays the most common content ratings on Netflix. Each bar represents the number of titles under a specific rating. Ratings such as TV-MA and TV-14 dominate the platform. This indicates that Netflix mainly targets mature and teenage audiences. The chart helps understand audience segmentation.

Histogram – Release Year

Illustrates the distribution of titles based on their release year. Most content is concentrated in recent years, especially after 2015. Older titles appear less frequently. The graph highlights Netflix's focus on newer releases. It also shows the overall trend of content production.

Histogram & Box Plot – Movie Duration

These graphs examine movie durations. The histogram shows that most movies last between 80 and 120 minutes. The box plot identifies the median, spread, and outliers. A few movies have unusually long durations. Together they summarize the duration distribution effectively.

Horizontal Bar Charts

The country chart identifies the top countries producing Netflix content, with the United States leading. The genre chart highlights the most common genres such as Drama and International Movies. These graphs compare categorical frequencies clearly. They help identify dominant production regions and popular content categories. They support content strategy analysis.

Pie Chart

Represents the proportion of Movies and TV Shows. Each slice shows the percentage contribution of a content type. Movies occupy the larger share of the chart. The visualization is useful for understanding overall composition. It provides an intuitive comparison.

Scatter Plot

Compares movie duration with release year. Each point represents a movie. The graph helps detect any relationship between duration and release period. Most durations remain within a similar range across years. No strong correlation is observed.

Line Plot

Shows the number of titles released over time. The trend rises sharply after 2015, indicating rapid content growth. Small fluctuations appear in certain years. The visualization highlights Netflix's expansion. It is useful for trend analysis.

Heatmap

Displays the relationship between content rating and content type. Darker colors indicate higher frequencies. It becomes easy to identify which ratings are common for Movies or TV Shows. The heatmap simplifies comparison across categories. It reveals concentration patterns.

Violin, Point & Strip Plots

These plots analyze duration and release year across different ratings and content types. The violin plot shows data distribution, the point plot presents average durations, and the strip plot displays individual observations. Together they reveal spread, central tendency, and variability. They provide deeper insights than simple bar charts.

Correlation Heatmap

Measures the correlation among numerical variables. Values close to 1 indicate strong positive relationships, while values near 0 show weak relationships. In this dataset, numerical features exhibit limited correlation. The heatmap provides a quick statistical overview. It supports feature understanding.

3D Scatter Plot, Treemap & Bubble Chart

These advanced visualizations explore multiple variables simultaneously. The 3D scatter plot compares release year, duration, and rating. The treemap shows the hierarchical distribution of content by country and type. The bubble chart highlights duration using bubble size and rating using color. Together they provide a comprehensive multivariate analysis.

Key Findings

- Movies dominate the Netflix catalog.
- Most titles were released after 2015.
- TV-MA and TV-14 are the most common ratings.
- The United States contributes the highest number of titles.
- Drama and International Movies are among the most popular genres.
- Most movies have durations between 80 and 120 minutes.

Conclusion

The EDA successfully transformed raw Netflix data into meaningful insights using data cleaning and visualization techniques. The findings reveal trends in content type, ratings, countries, genres, release years, and durations. These insights can assist decision-makers in understanding audience preferences and planning future content strategies.